

KUDAT, Malaysia

WMO: 964770

Lat: 6.917N

Lon: 116.833E

Elev: 5

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	22.9	20.8	15.5	27.4	21.8	16.4	26.5	7.6	27.8	7.0	27.9	1.2	230	0.348

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	7.0	33.8	26.8	33.1	26.8	32.9	26.7	28.2	31.9	27.7	31.4	27.5	31.0	4.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.7	26.9	22.6	30.6	26.2	21.7	29.7	90.5	32.1	88.5	32.0	87.2	31.8	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	6.1	5.4	21.1	34.8	0.9	0.8	20.5	35.3	20.0	35.7	19.5	36.2	18.9	36.7		
			20.7	29.0	0.7	0.8	20.2	29.6	19.8	30.0	19.4	30.5	18.9	31.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	27.0	27.4	27.9	28.6	28.7	28.2	27.8	27.9	27.8	27.6	27.4	27.2
	DBStd	1.14	1.11	1.17	1.10	1.14	1.03	0.95	0.96	0.95	0.94	0.97	0.91	1.10
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6494	528	487	554	558	579	546	551	556	533	546	523	534
	CDD18.3	3452	269	253	296	308	321	296	293	297	283	288	273	276
	CDH23.3	36463	2607	2554	3202	3651	3695	3225	3008	3176	2941	2928	2748	2728
	CDH26.7	13081	636	673	1077	1536	1597	1328	1146	1284	1103	1031	898	773
Wind	WSAvg	2.4	2.7	2.9	2.6	2.2	2.1	2.1	2.3	2.4	2.3	2.4	2.0	2.3
Precipitation	PrecAvg	2460	418	214	171	77	136	141	127	120	117	186	267	485
	PrecMax	4014	1005	694	511	151	274	225	295	198	198	360	500	909
	PrecMin	1508	123	42	36	5	32	17	18	27	23	102	55	126
	PrecStd	631	240	174	134	43	65	54	63	51	49	68	107	211
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	32.1	32.8	34.1	34.2	34.1	33.2	34.0	33.2	33.0	32.8	32.2
		MCWB	26.7	26.7	27.1	27.8	27.4	26.5	26.2	25.8	26.2	26.8	27.3	26.9
	2%	DB	30.9	31.1	32.0	33.1	33.2	33.2	32.9	33.1	32.8	32.1	31.9	31.2
		MCWB	26.6	26.5	26.7	27.4	27.2	26.5	26.3	26.1	26.3	26.6	26.8	26.8
	5%	DB	30.1	30.2	31.2	32.8	32.9	32.8	32.1	32.2	32.0	31.8	31.1	30.8
		MCWB	26.2	26.1	26.6	27.2	27.1	26.5	26.1	26.1	26.2	26.5	26.7	26.7
	10%	DB	29.2	29.9	30.8	31.9	32.1	31.9	31.2	31.9	31.2	31.0	30.2	30.0
		MCWB	25.8	25.9	26.5	26.9	26.9	26.6	26.2	26.1	26.3	26.6	26.4	26.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.2	28.2	28.6	28.7	29.1	27.9	28.0	27.7	27.9	27.8	27.7
		MCDB	30.0	30.4	31.1	32.7	32.5	31.7	31.9	32.0	31.7	31.6	30.9	30.6
	2%	WB	26.8	26.8	27.3	28.0	28.0	28.0	27.2	27.3	27.2	27.3	27.2	27.2
		MCDB	29.4	30.0	30.5	31.9	31.9	31.5	31.0	31.2	30.9	30.7	30.2	29.8
	5%	WB	26.5	26.5	26.9	27.5	27.6	27.3	26.7	26.8	26.7	26.9	26.9	26.8
		MCDB	29.0	29.4	30.0	31.2	31.3	30.9	30.4	30.5	30.2	30.2	29.8	29.3
	10%	WB	26.2	26.2	26.6	27.1	27.2	26.9	26.5	26.5	26.5	26.6	26.6	26.6
		MCDB	28.5	28.7	29.5	30.5	30.8	30.4	30.0	30.2	29.9	29.8	29.4	28.9
Mean Daily Temperature Range	5% DB	MDBR	4.9	4.9	5.9	7.0	7.6	7.9	7.5	8.0	7.5	6.9	6.4	5.3
		MCDBR	5.7	5.6	6.6	7.8	8.2	9.0	8.4	8.7	8.4	7.8	7.3	6.5
		MCWBR	2.8	2.6	2.9	3.2	3.3	3.7	3.5	3.6	3.4	3.3	3.3	3.1
	5% WB	MCDBR	5.2	5.3	6.1	7.2	7.7	8.1	7.8	8.3	7.8	7.2	6.6	5.7
MCWBR		2.7	2.7	2.9	3.2	3.5	3.9	3.6	3.6	3.5	3.4	3.2	2.9	
Clear-Sky Solar Irradiance	taub	0.430	0.426	0.424	0.424	0.411	0.416	0.418	0.445	0.440	0.444	0.419	0.423	
	taud	2.459	2.444	2.456	2.447	2.506	2.505	2.497	2.394	2.404	2.394	2.482	2.480	
	Ebn at Noon	887	899	900	885	878	863	865	856	873	871	891	886	
	Edh at Noon	114	118	118	117	108	106	108	122	122	123	110	110	
All-Sky Solar Radiation	RadAvg	4.42	5.29	5.76	6.13	5.65	5.10	5.03	5.20	5.10	4.82	4.51	4.19	
	RadStd	0.58	0.63	0.66	0.44	0.39	0.35	0.31	0.34	0.40	0.33	0.39	0.50	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	-0.37	-0.37	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+0.36	+0.39	+0.30	N/A	N/A	+99	+98

Nomenclature: See separate page